

System of Linear Equations

Definition The system of n equations in n unknowns $x_1, x_2, \dots, x_n$ of the form

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\&\vdots \\a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n &= b_n\end{aligned}$$

is known as a system of simultaneous (non-homogenous) linear equations, where $b_1, b_2, \dots, b_n$ are constants.

Definition: By the solution of this system of equations we mean a set of values of $x_1, x_2, \dots, x_n$ which satisfies simultaneously these n equations. When a system of equations has one or more solutions, it is said to be consistent, otherwise the system of equations is said to be inconsistent.

This system of equations can also be written in the matrix form as: $AX = B$

$$\text{where } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

so that, A, X, B are matrices of order $n \times n, n \times 1, n \times 1$, respectively, and the matrix A is known as coefficient matrix.

CRITERION OF CONSISTENCY

If there is a system of n linear equations in n variables given by $AX = B$ and A is non-singular, then the system has unique solution given by $X = A^{-1}B$.

The following are criteria of consistency or inconsistency of a system of linear equations given $AX = B$, where A is square matrix.

- (i) If $|A| \neq 0$, then system is consistent and has a unique solution given by $X = A^{-1}B$.
- (ii) If $|A| = 0$, the system of equations has either no solution or an infinite number of solutions.
 - (a) If $(\text{adj } A)B \neq 0$, the system has no solution and is therefore inconsistent.
 - (b) If $(\text{adj } A)B = 0$, the system is consistent and has infinitely many solutions.

METHODS OF SOLVING NON-HOMOGENEOUS LINEAR EQUATIONS

The solution of linear non-homogeneous simultaneous equations can be obtained by applying following methods:

1. Matrix Inverse Method
2. Gauss Elimination Method
3. Cramer's Rule

1. Inverse matrix method:

If $|A| \neq 0$, then A is non-singular and hence its inverse, A^{-1} exists and unique. Take the system of equations $AX = B$. Pre-multiplying both sides by A^{-1} , we have

$$\begin{aligned} A^{-1}(AX) &= A^{-1}B \\ \text{or, } IX &= A^{-1}B \quad (\because A^{-1}A = I) \\ \text{or, } X &= A^{-1}B = \frac{\text{adj } A}{|A|} \times B \end{aligned}$$

which gives the solution of the system of simultaneous (non-homogenous) equations when the number of variables are equal to the number of equations.

2. Gauss elimination method (or Triangular form reduction method)

This method is also known as **pivotal reduction method**. According to this method the co-efficient matrix $|a_{ij}|$ for a system of simultaneous equations (homogeneous or non-homogeneous) is reduced in such a way that $a_{ij} = 0$ for $j > i$.

This method can be used even if the co-efficient matrix is singular or non-singular, the number of equations equals to the number of unknowns or otherwise.

CRAMER'S RULE

One of the applications of the determinants is to solve a system of linear equations in which number of variables are equal to the number of equations and the co-efficient matrix of the system of equations is non-singular. This method enables us to determine solution directly without computing the inverse of the matrix.

Let us consider the following linear equations:

$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned} \quad \text{or, } \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Let D be the determinant of the co-efficient of the variables x and y such that

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

Further, let $D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$ and $D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$ be the determinant obtained from D by replacing the first column by the elements c_1, c_2 and by replacing the second column by the elements c_1, c_2 respectively. Thus the values of x and y can be expressed in the form of determinant as:

$$\frac{x}{D_x} = \frac{y}{D_y} = \frac{1}{D}, \text{ provided } D \neq 0.$$

$$\therefore x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}, \quad z = \frac{D_z}{D}, \text{ if } D \neq 0.$$

The Cramer's rule for solving a system of linear equations can also be extended for equations involving more than two variables by using following general formula:

Cramer's formula: $x_j = \frac{D_j}{D}, j = 1, 2, \dots, n.$

Where the determinant D_i can be obtained from D (determinant of the coefficients of unknowns) by replacing i th column of D by the column of right hand side constant terms.

Problem-1:

Solve the following system of equations using Cramer's rule
 $x+y = 11, 2x-6y=0.$

Solution: The determinant of the coefficients of x and y is:

$$D = \begin{vmatrix} 1 & 1 \\ 2 & -6 \end{vmatrix} = -6 - 2 = -8 (\neq 0).$$

To find D_x , we change first column of D by R.H.S. terms 11, 0. Then

$$D_x = \begin{vmatrix} 11 & 1 \\ 0 & -6 \end{vmatrix} = -66.$$

Similarly to find D_y , we change first second column of D by R.H.S. terms 11, 0.

Then
$$D_y = \begin{vmatrix} 1 & 11 \\ 2 & 0 \end{vmatrix} = -22.$$

Hence,
$$x = \frac{D_x}{D} = \frac{-66}{-8} = \frac{33}{4}; \quad y = \frac{D_y}{D} = \frac{-22}{-8} = \frac{11}{4}.$$

Problem-2:

Solve the following system of equations using Cramer's method.

$$x+y+z=6; \quad x-y+z=2; \quad 2x+y-z=1$$

Solution: The determinant of the coefficients of x, y and z is given by

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix} = (1-1) - 1(-1-2) + 1(1+2) = 6 (\neq 0).$$

The solution is given by $\frac{x}{D_x} = \frac{y}{D_y} = \frac{z}{D_z} = \frac{1}{D}$

Where $D_x = \begin{vmatrix} 6 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 6 \begin{vmatrix} -1 & 1 \\ 1 & -1 \end{vmatrix} - \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} = 6$

$$D_y = \begin{vmatrix} 1 & 6 & 1 \\ 1 & 2 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 1 \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} - 6 \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 12$$

$$D_z = \begin{vmatrix} 1 & 1 & 6 \\ 1 & -1 & 2 \\ 2 & 1 & 1 \end{vmatrix} = 1 \begin{vmatrix} -1 & 2 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} + 6 \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 18$$

Thus, we have $\frac{x}{6} = \frac{y}{12} = \frac{z}{18} = \frac{1}{6}$ or $x = 1, y = 2, z = 3$.

Problem-3:

Use matrix method to examine the following system of equations for consistency or inconsistency : $2x-3y=3$, and $4x-y=11$.

Solution: The given equations in the matrix form can be written as:

$$\begin{bmatrix} 2 & -3 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 11 \end{bmatrix} \text{ or, } Ax = B$$

$$\text{where } A = \begin{bmatrix} 2 & -3 \\ 4 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \end{bmatrix}, B = \begin{bmatrix} 3 \\ 11 \end{bmatrix}$$

Since $|A| = \begin{vmatrix} 2 & -3 \\ 4 & -1 \end{vmatrix} = 10 (\neq 0)$, i.e. A is non-singular. A^{-1} exists and the system is consistent and has a unique solution given by

$$AX = B \quad \text{or,} \quad X = A^{-1} B \quad \dots(i)$$

To determine A^{-1} first we determine matrix of co-factors of each element of matrix A as follows:

$$A_{11} = (-1)^{1+1} (-1) = -1 \quad A_{12} = (-1)^{1+2} 4 = -4$$

$$A_{21} = (1)^{2+1} (-3) = 3 \quad A_{22} = (-1)^{2+2} 2 = 2$$

$$\therefore \text{adj } A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}^T = \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ -4 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj } A}{|A|} = \frac{1}{10} \begin{bmatrix} -1 & 3 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} -1/10 & 3/10 \\ -4/10 & 2/10 \end{bmatrix} \quad \dots(ii)$$

From (i) and (ii), we get

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1/10 & 3/10 \\ -4/10 & 2/10 \end{bmatrix} \begin{bmatrix} 3 \\ 11 \end{bmatrix} = \begin{bmatrix} -3/10 + 33/10 \\ -12/10 + 22/10 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Hence, the unique solution is given by; $x = 3$ and $y = 1$.

Problem-4:

By matrix method solve the following equations:

$$5x - 6y + 4z = 15$$

$$7x + 4y - 3z = 19$$

$$2x + y + 6z = 46.$$

Solution: The given system of equations may be represented in the matrix notations as:

$$\begin{bmatrix} 5 & -6 & 4 \\ 7 & 4 & -3 \\ 2 & 1 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 19 \\ 46 \end{bmatrix}$$

$$\text{or,} \quad AX=B \quad \text{or} \quad X=A^{-1} B$$

$$\text{Since} \quad |A| = \begin{vmatrix} 5 & -6 & 4 \\ 7 & 4 & -3 \\ 2 & 1 & 6 \end{vmatrix} = 5 \begin{vmatrix} 4 & -3 \\ 1 & 6 \end{vmatrix} + 6 \begin{vmatrix} 7 & -3 \\ 2 & 6 \end{vmatrix} + 4 \begin{vmatrix} 7 & 4 \\ 2 & 1 \end{vmatrix} = 135 + 288 - 4 = 419 \neq 0$$

Hence, A is non-singular and A^{-1} exists. The system has the unique solution.

The matrix of co-factors of elements of A is given by

$$\begin{aligned} A_{11} &= 27, & A_{12} &= -48 & A_{13} &= -1 \\ A_{21} &= 40, & A_{22} &= 22, & A_{23} &= -17 \\ A_{31} &= 2, & A_{32} &= 43, & A_{33} &= 62. \end{aligned}$$

$$\text{adj } A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} = \begin{bmatrix} 27 & 40 & 2 \\ -48 & 22 & 43 \\ -1 & -17 & 62 \end{bmatrix}$$

$$\text{And } A^{-1} = \frac{\text{Adj } A}{A} = \frac{1}{419} \begin{bmatrix} 27 & 40 & 2 \\ -48 & 22 & 43 \\ -1 & -17 & 62 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{419} \begin{bmatrix} 27 & 40 & 2 \\ -48 & 22 & 43 \\ -1 & -17 & 62 \end{bmatrix} \begin{bmatrix} 15 \\ 19 \\ 46 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}$$

Hence, the unique solution of the system is: $x = 3$, $y = 4$, $z = 6$.

Problem-5:

Solve the following system of equation using Gauss elimination method

$$2x - y + 3z = 9; x + y + z = 6 \text{ and } x - y + z = 2.$$

Solution: The matrix equation of given equations can be written as

$$\begin{bmatrix} 2 & -1 & 3 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ 2 \end{bmatrix} \text{ or, } AX = B,$$

Applying row operations: $R'_2 \rightarrow 2R_2 - R_1$ and $R'_3 \rightarrow 2R_3 - R_1$, we get

$$\begin{bmatrix} 2 & -1 & 3 \\ 0 & 3 & -1 \\ 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 3 \\ -5 \end{bmatrix}; R'_3 \rightarrow R_3 + 3R_2$$

$$\begin{bmatrix} 2 & -1 & 3 \\ 0 & 3 & -1 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 3 \\ -12 \end{bmatrix}$$

The given system of equation can be rewritten as:

$$2x - y + 3z = 9 \quad \dots(i)$$

$$3y - z = 3 \quad \dots(ii)$$

$$-4z = -12 \quad \dots(iii)$$

From (iii), $z = 3$.

Putting $z = 3$ in (ii) $3y - 3 = 3$ or, $y = 2$.

Also on putting values of y, z , in (i) we get, $2x - 2 + 9 = 9$ or, $x = 1$.

Hence, the unique solution is: $x = 1, y = 2, z = 3$.

Problem-6:

Solve the system of equations by using Gauss elimination method:

$$x + y + z = 9$$

$$2x + 5y + 7z = 52$$

$$2x + y - z = 0.$$

Solution: The system of equations can also be represented as:

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$$

Applying elementary row operations: $R_2 \rightarrow R_2 - 2R_1$ and $R_3 \rightarrow R_3 - 2R_1$, we get

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 5 \\ 0 & -1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 34 \\ -18 \end{bmatrix}; R'_2 \rightarrow R_2 + 3R_3$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 0 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 9 \\ -20 \\ -18 \end{bmatrix}$$

Rewriting equations as follows:

$$x + y + z = 9$$

$$-4z = -20$$

$$-y - 3z = -18$$

Solving these equations, we get $x = 1, y = 3$, and $z = 5$.

Exercise

1. Solve the following system of linear equations by using i) Cramer's rule ii) Matrix Method and iii) Gauss Elimination method.

(a) $x - y + 2z = 9$; $2x - 5y + 3z = 18$; $6x + 7y + 10z = 35$

(b) $3x + 3y - z = 11$; $2x - y + 2z = 9$; $4x + 3y + 2z = 25$

(c) $x + y - z = -3$; $2x + 3y + z = 2$; $8y + 3z = 1$

(d) $x + y + z + 1 = 0$; $x + 2y + 3z + 4 = 0$; $x + 3y + 4z + 6 = 0$

(e) $2x + y + z = 7$; $3x - y - z = -2$; $x + 2y - 3z = -4$

(f) $2x - 3y + z = 7$; $2x + y - z = 1$; $4x + 3z = -11$.

2. Solve the following system of linear equations by using i) Cramer's rule ii) Matrix Method and iii) Gauss Elimination method.

(a) $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$

(b) $2x + 4y + 5z = 18$, $2x - y + 8z = 13$, $5x - 2y + 7z = 20$

(c) $2x - y + 3z = 9$, $x + 3y - z = 4$, $3x + 2y + z = 10$

3. Solve the following system of linear equations by using i) Cramer's rule ii) Matrix Method and iii) Gauss Elimination method.

(a) $2x - y + 3z = 9$, $-x + 2y + z = 6$, $3x + y - 4z = -7$

(b) $2x + 3y - z = 9$, $x + y + z = 9$, $3x - y - z = -1$

(c) $9x + 7y + 3z = 6$, $5x + y + 4z = 1$, $6x + 8y + 2z = 4$.